Rational and Radical Expressions

Assessment

Name _____ Date _____ Period _____

Show all work. Write your final answer on each answer line. Partial credit may be awarded for correct steps.

Algebra 2

1. Given expression = $\frac{x^2 - 25}{x - 5}$. Find the simplified form (state any restrictions).

Answer _____

2. Given expression = $\frac{x^2 - x}{x}$. Find the simplified form (state any restrictions).

Answer _____

3. Given expression = $\frac{x^2 - 1}{x - 3} \div \frac{x + 1}{x - 3}$. Find the simplified product/quotient.

Answer _____

4. Given expression = $\frac{x^2 - 4}{x + 1} \cdot \frac{x + 1}{x + 2}$. Find the simplified product/quotient.

Answer _____

5. Given expression = $\sqrt{896}$. Find the simplified form.

Answer _____

6. Given expression = $\sqrt{32}$. Find the simplified form.

Answer _____

7. Given expression = $\sqrt{y^5}$. Find the the same expression using a rational exponent.

Answer _____

8. Given expression = $\sqrt[3]{x^4}$. Find the the same expression using a rational exponent.

Answer _____

9. Given equation = $\sqrt{x + 16} = x - 4$. Find the the solution set, checking every candidate for extraneous solutions.

Answer _____

10. Given equation = $\sqrt{x - 1} = x - 3$. Find the the solution set, checking every candidate for extraneous solutions.

Answer _____

ANSWER KEY

DO NOT DISTRIBUTE, TEACHER USE ONLY

Rational and Radical Expressions

1. $x + 5, x \neq 5$

2. $x - 1, x \neq 0$

3. $x - 1$

4. $x - 2$

5. $8\sqrt{14}$

6. $4\sqrt{2}$

7. $y^{(5/2)}$

8. $x^{(4/3)}$

9. {9}

10. {5}